

Group Topic: PH Staple Food Price Analytics — Association Rule Mining (SDG 2) **Lab No.** Chapter 9 **Git Repo/ Colab Link:** https://github.com/jaysonbuenosaires/Buenosaires_Elec2 **Date:** May 16, 2026

Objectives

- To implement the FP-Growth algorithm to bypass the severe memory limitations encountered during the Apriori testing phase.
- To successfully mine the complete, un-sampled dataset of 46,591 Co-Occurrence and 38,053 Price Spike transaction baskets.
- To apply highly sensitive minimum support thresholds to capture nuanced, lower-frequency agricultural market behaviors.
- To generate the definitive set of association rules for the project's final economic analysis.

Tasks Completed Lead ML Engineer

- **Tasks:** FP-Growth implementation, full-dataset mining, and algorithmic scaling.
- **Description:**
 - Configured and deployed the `fpgrowth` function from the `mlxtend` library to process the full boolean-encoded transaction datasets.
 - Bypassed the memory bottlenecks that crashed Apriori by utilizing the algorithm's FP-tree compression architecture, which entirely eliminates the need for dense candidate generation.
 - Lowered the minimum support thresholds significantly to capture deeper insights: set to 0.05 for Co-Occurrence and 0.10 for Price Spikes.
 - Successfully generated an immense 4,520,169 co-occurrence rules (in 48.97 seconds) and 1,272 price spike rules (in 3.46 seconds), proving the model's scalability.
- **Screenshot/Code Snippet:**

Python

```
# Running FP-Growth on full Price Spike transactions with max_len=4
spike_freq_fp = fpgrowth(spike_enc, min_support=MIN_SUP_SPIKE,
                        use_colnames=True, max_len=4)

spike_rules_fp = association_rules(
    spike_freq_fp, num_itemsets=len(spike_freq_fp),
    metric='confidence', min_threshold=MIN_CONF)
```

Tasks to be Completed

- Cross-validate the top 50 high-lift FP-Growth rules against external agricultural news or inflation reports from the 2018–2021 period to verify their real-world accuracy.
- Draft the algorithmic methodology section for the final manuscript detailing the mathematical advantages of the FP-tree structure over Apriori's candidate generation.

Results and Analysis

1. **Algorithmic Superiority & Scalability:** The FP-Growth implementation was the pivotal technical achievement of the pipeline. While Apriori required the data to be downsampled to 15,000 rows just to prevent system crashes, FP-Growth effortlessly processed the entire array of transactions. It discovered 508,474 frequent itemsets for co-occurrence compared to Apriori's 27,251, proving it is the only viable option for large agricultural datasets.
2. **Deep Supply Chain Bundling (Co-Occurrence):** Because FP-Growth allowed for a lower support threshold (0.05), it uncovered highly specific inventory combinations. The model proved that premium items are tightly bundled with baseline aromatics: a market selling "Meat (pork, with fat)" and "Rice (special)" has a massive Lift of 12.03 for also stocking "Garlic," mapping out consistent retail inventory behaviors.
3. **Predictive Inflation Clusters (Price Spikes):** The unconstrained algorithm successfully mapped complex inflationary contagion involving core proteins and staples. It generated 1,272 highly relevant spike rules. A key finding was that simultaneous price shocks in "Eggs" and "Pork" heavily predict a subsequent/concurrent spike in both "Milkfish" and "Rice (regular, milled)" with a high Lift of 6.56.

Challenges and Solutions

- **Challenge:** Processing the massive volume of generated rules (over 4.5 million for Co-occurrence) threatened to create downstream memory issues for pandas DataFrame operations during the evaluation phase, creating a "needle in a haystack" scenario.
- **Solution:** I optimized the rule extraction phase by applying strict `max_len=4` parameters to prevent the generation of overly complex, uninterpretable 5+ item rules. I then aggressively sorted and filtered the output exclusively by the `lift` metric immediately after generation to ensure a manageable and highly relevant handoff to the Data Analyst.

Conclusion The implementation of the FP-Growth algorithm successfully solved the project's primary technical hurdle: achieving big data scalability. By leveraging the FP-tree structure, the model mined the entirety of the transaction data without memory failure. This comprehensive execution yielded deep, actionable intelligence regarding agricultural supply chain bundling and predictive indicators for regional food inflation, fully satisfying the core analytical requirements of the ML phase.